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- Ignores syntactic boundaries during denoising
- Fails to capture long-range hierarchical dependencies
- Treats all tokens equally — but they're not!

*"For code, noise should not be purely stochastic — structural priors enable DLLMs to learn program-level dependencies."*

TreeDiff parses code into an **Abstract Syntax Tree (AST)** and uses it to guide which tokens to mask — targeting structurally critical nodes.



| Category     | Examples           | Weight | Purpose            |
|--------------|--------------------|--------|--------------------|
| Skeletal     | Imports, Constants | 0.15   | Preserve skeleton  |
| Data Flow    | Assigns, Calls     | 0.42   | Moderate priority  |
| Conditionals | If, elif, else     | 0.58   | Core logic         |
| Control Flow | for, while, return | 0.60   | 🚩 Highest priority |

TreeDiff maintains stable performance ( $-0.1\%$ ) as  $T$  increases  $256 \rightarrow 512$ , while baselines degrade by  $\sim 2.1\%$

**TreeDiff:** Establishes stack-based control flow (`stack.pop()`) early. Correctly captures recursive nesting

## Key Contributions

- **First AST-aware masking** for DLLMs in code generation
- **13.3% relative improvement** on HumanEval+ vs random masking
- Trained on **150K** long code reasoning samples
- Negligible overhead: AST parsing is  $O(L)$

TreeDiff addresses a fundamental limitation: **random masking ignores code structure**. By aligning corruption with AST hierarchy, the model learns programming language compositionality, reconstructing programs that respect grammatical boundaries and capture long-range dependencies.

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**Setup:** LLaDA-8B backbone · 8× NVIDIA A100 · LoRA ( $r=64$ ,  $\alpha=128$ ) · 4096 max tokens · LR  $5 \times 10^{-5}$  · Benchmarks: HumanEval/+, MBPP/+